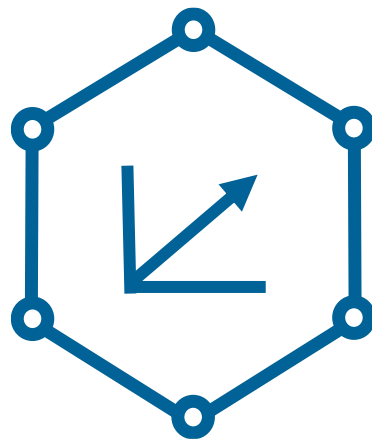


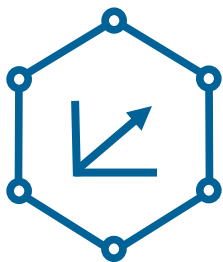


Knowledge Graph Embeddings

Emanuel Sallinger

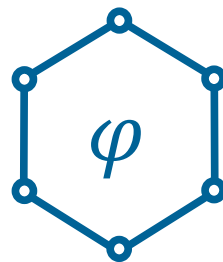


Representations



KG Embeddings

Widely-applied, large family of ML models



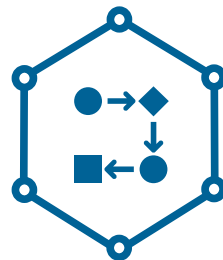
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



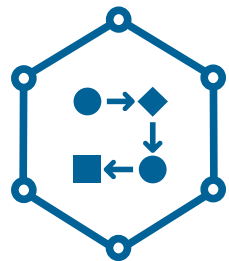
Graph Neural Networks

Using the KG structure as a neural network.



Data Models

Overview of data models in different communities



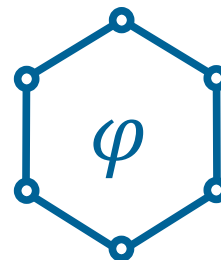
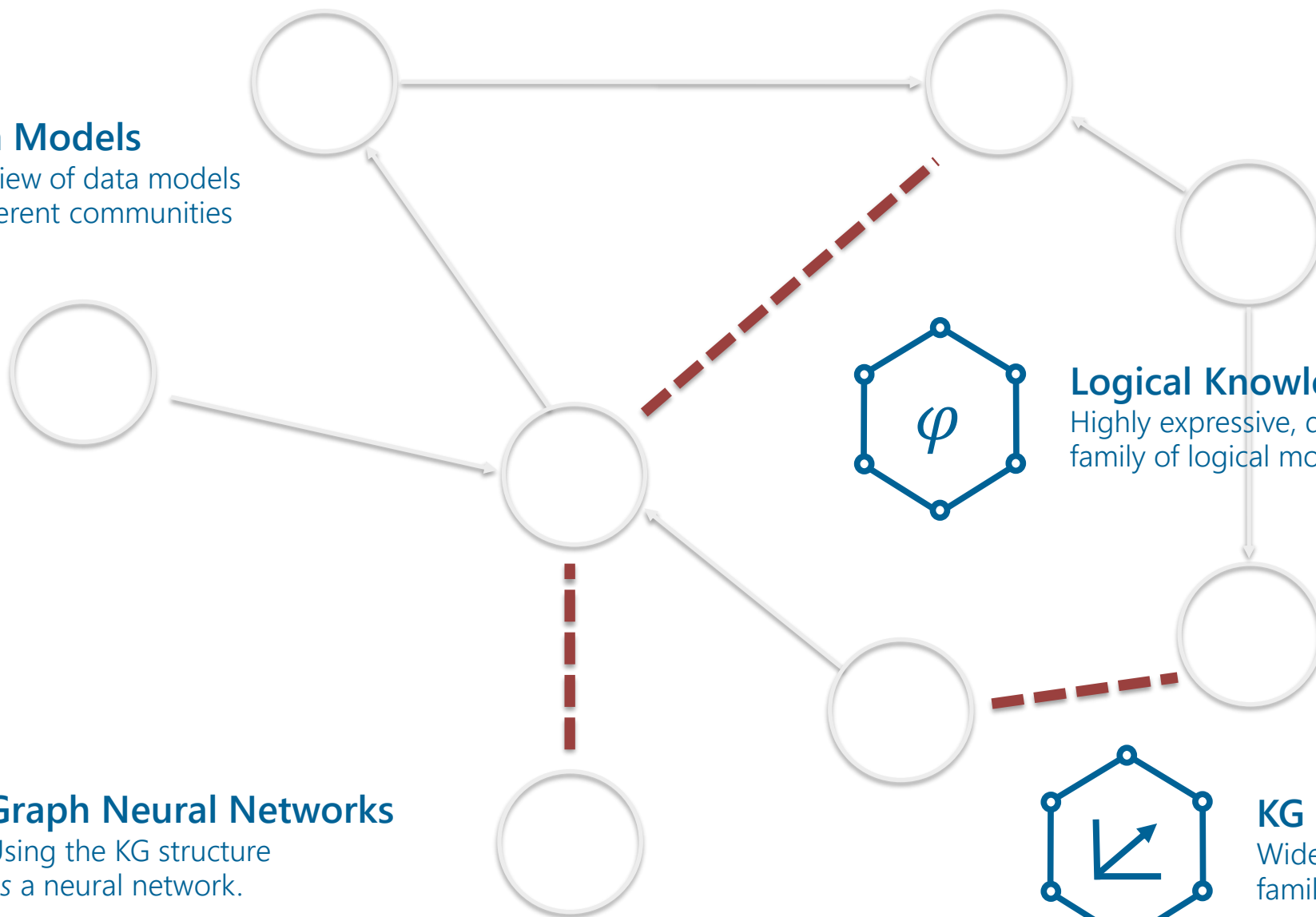
Data Models

Overview of data models
in different communities



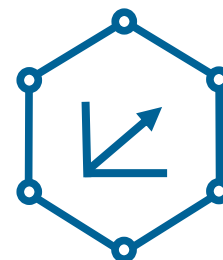
Graph Neural Networks

Using the KG structure
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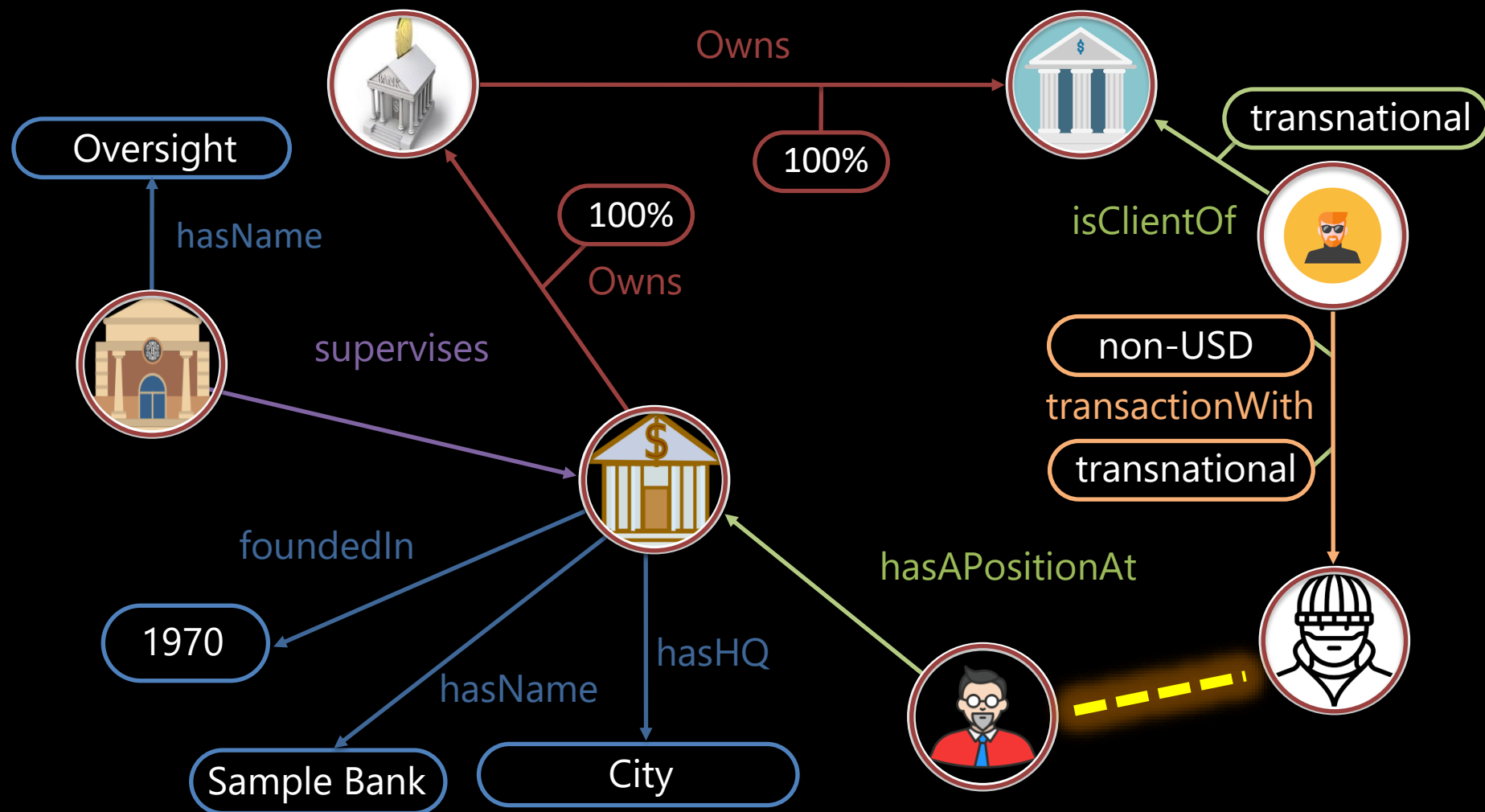
Logical Knowledge in KGs

Highly expressive, diverse
family of logical models.



KG Embeddings

Widely-applied, large
family of ML models

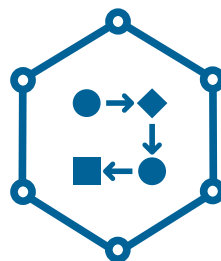




Representations



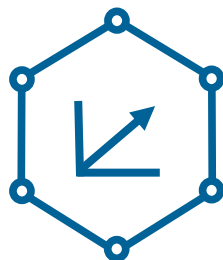
Graph



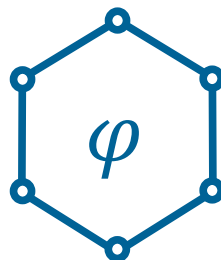
Knowledge Graph
Data Models



Knowledge



Knowledge Graph
Embeddings



Logical
Knowledge



Graph Neural
Networks

What is an Embedding?

$$e: \mathbb{S}_1 \rightarrow \mathbb{S}_2$$

What is an Embedding?

$$e: \mathbb{S}_1 \rightarrow \mathbb{S}_2$$

In mathematics often: $\mathbb{S}_1 \supset \mathbb{S}_2$,
 e is injective and *structure preserving*



Many Embeddings in Computer Science

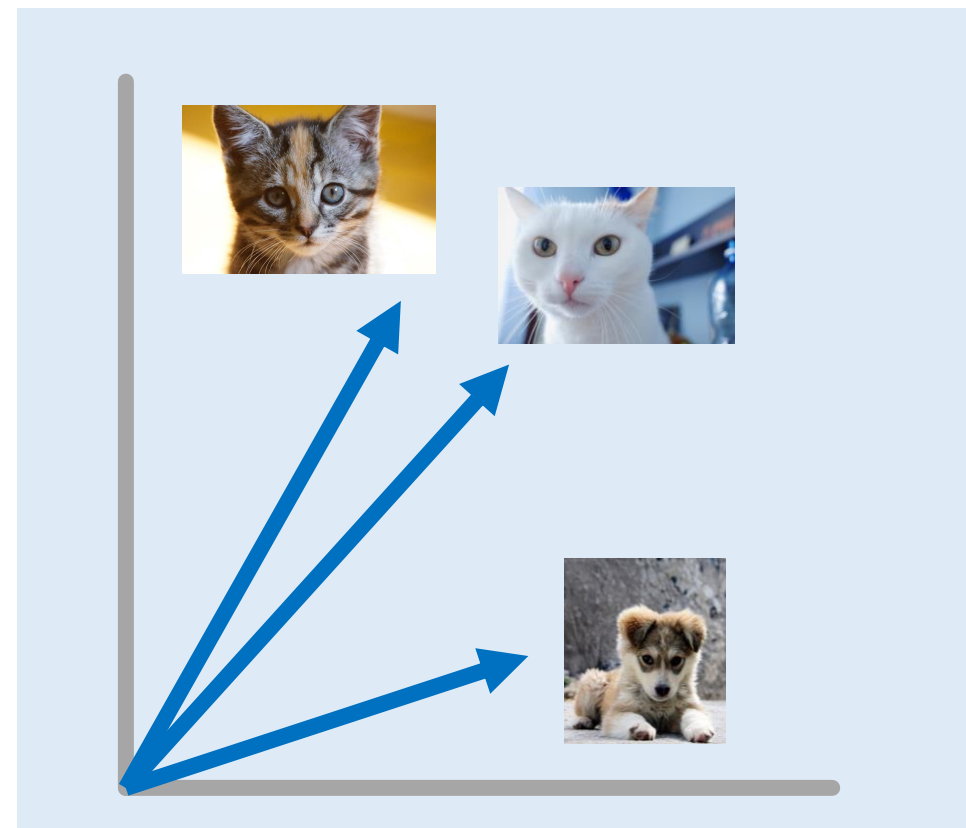
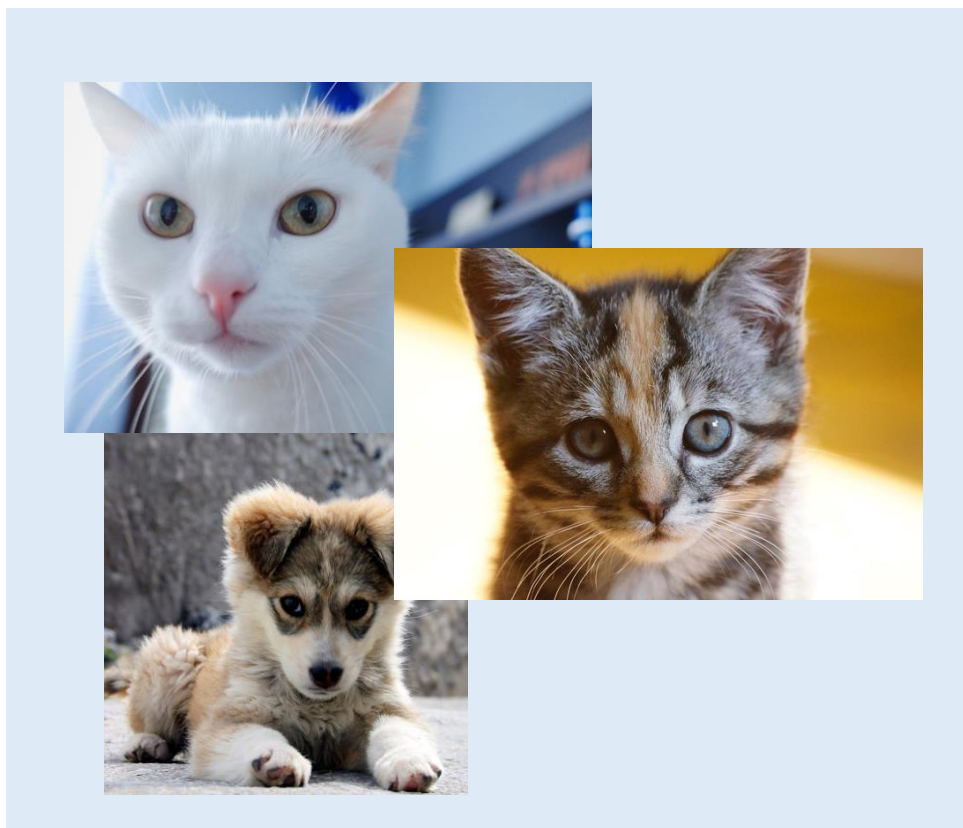
$$e: \mathbb{D} \rightarrow \mathbb{R}^d$$

where:

- $\mathbb{D}$ is some datatype
- $\mathbb{R}^d$ are d -dimensional vectors



$$e: \text{Images} \rightarrow \mathbb{R}^2$$





$$e: \text{Words} \rightarrow \mathbb{R}^2$$

Cat

Kitten

Dog



Cat

Kitten

Dog



Source: https://www.tensorflow.org/tutorials/text/word_embeddings

RSVP for your your local TensorFlow Everywhere event today!

Find an event

TensorFlow > Learn > TensorFlow Core > Tutorials

☆☆☆☆☆

Word embeddings

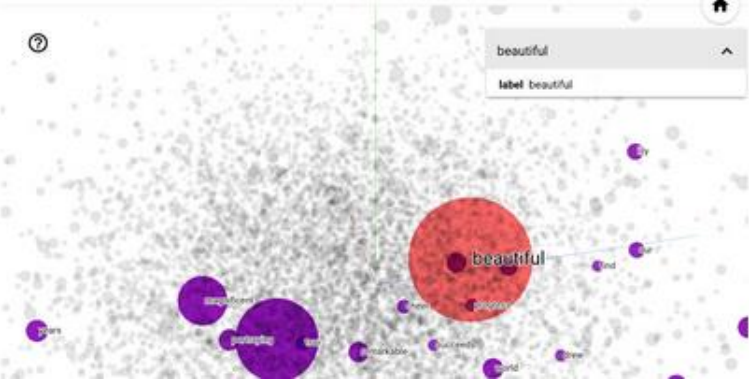
 Run in Google Colab

 View source on GitHub

 Download notebook

This tutorial contains an introduction to word embeddings. You will train your own word embeddings using a simple Keras model for a sentiment classification task, and then visualize them in the [Embedding Projector](#) (shown in the image below).

Points: 10000 | Dimension: 16 | Selected 101 points





Many Embeddings in Computer Science

$$e: \mathbb{D} \rightarrow \mathbb{R}^{d_1, d_2}$$

where:

- $\mathbb{D}$ is some datatype
- $\mathbb{R}^d$ are d_1, d_2 -dimensional matrices



Many Embeddings in Computer Science

$$e: \mathbb{D} \rightarrow \mathbb{C}^d$$

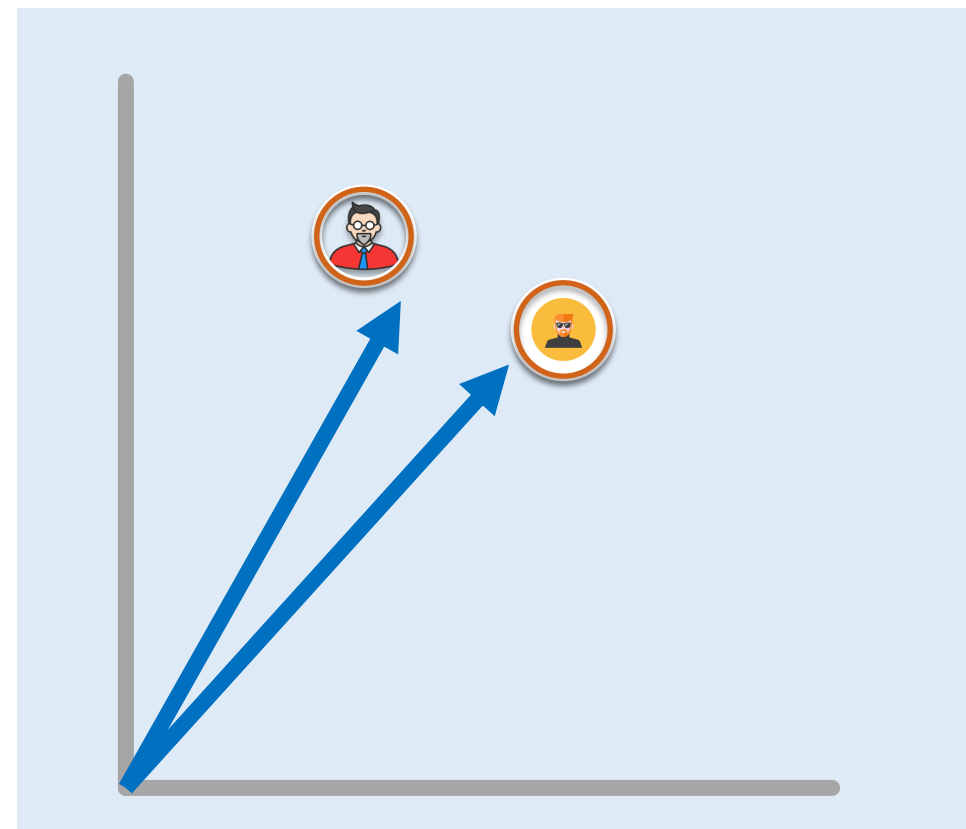
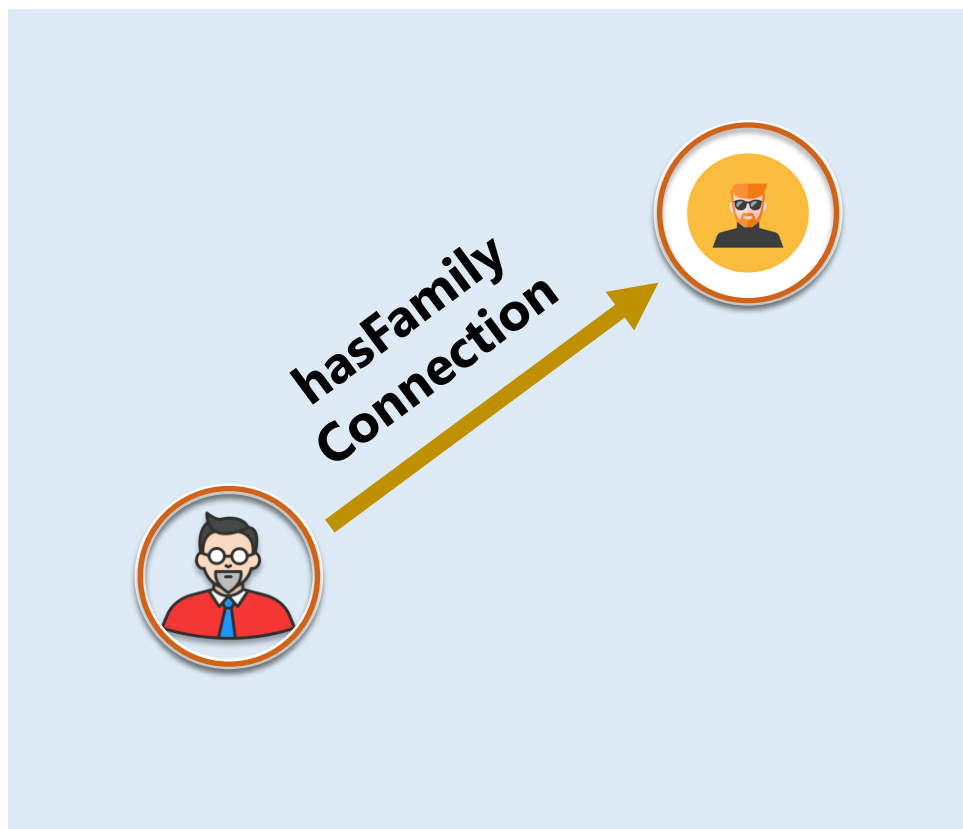
How does this help us?

$$KG = (D, K; g)$$

$$g: \mathbb{D} \rightarrow (\mathbb{V} \times \mathbb{E})$$



$$e: V \rightarrow \mathbb{R}^2$$





Source: <https://snap.stanford.edu/node2vec/>

By Jure Leskovec

STANFORD
UNIVERSITY



- ▶ SNAP Code
- ▶ SNAP Datasets
- ▶ What's new
- Papers
- Links
- About

node2vec: Scalable Feature Learning for Networks

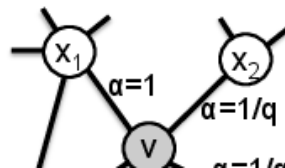
node2vec is an algorithmic framework for representational learning on graphs. Given any graph, it can learn continuous feature representations for the nodes, which can then be used for various downstream machine learning tasks.

- [Motivation](#)
- [Code](#)
- [Datasets](#)
- [Contributors](#)
- [References](#)

Motivation

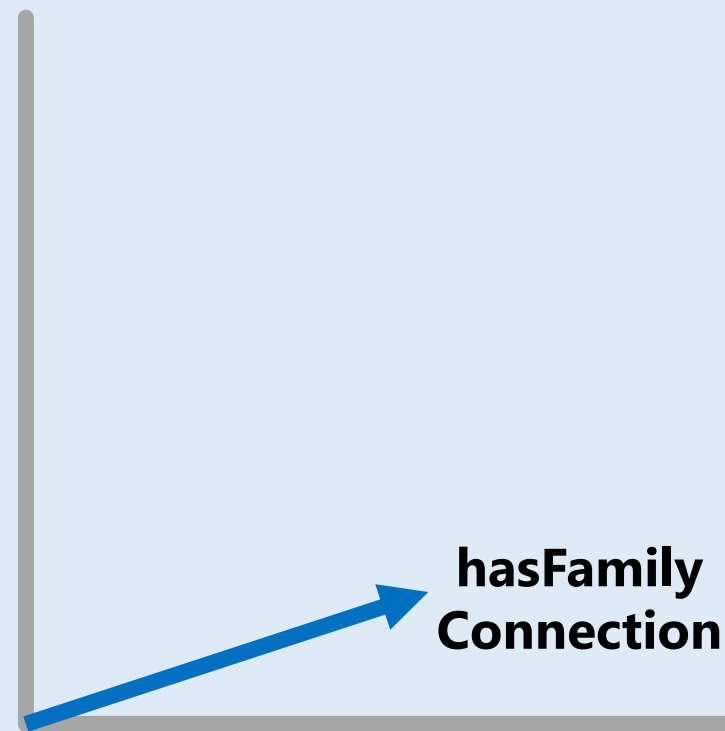
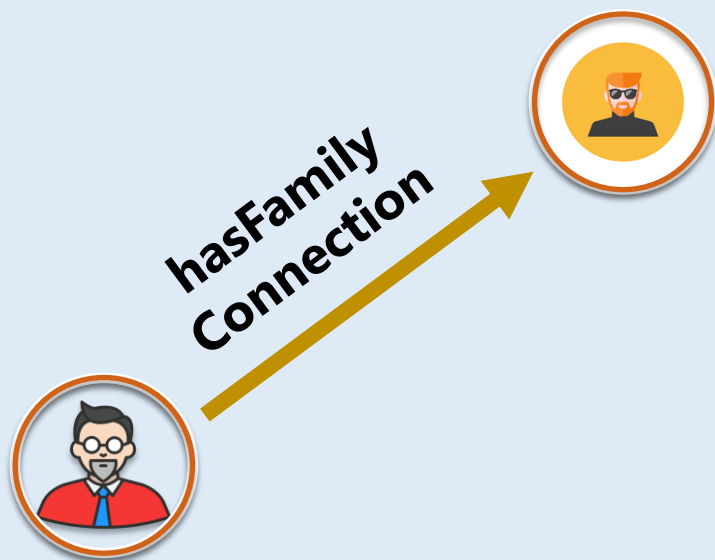
Learning useful representations from highly structured objects such as graphs is useful for a variety of machine learning applications. Besides reducing the engineering effort, these representations can lead to greater predictive power.

The *node2vec* framework learns low-dimensional representations for nodes in a graph by optimizing a neighborhood preserving objective. The objective is flexible, and the algorithm accommodates for various definitions of network neighborhoods by simulating biased random walks. Specifically, it provides a way of balancing the exploration-exploitation tradeoff that in turn leads to representations obeying a spectrum of equivalences from homophily to structural equivalence.



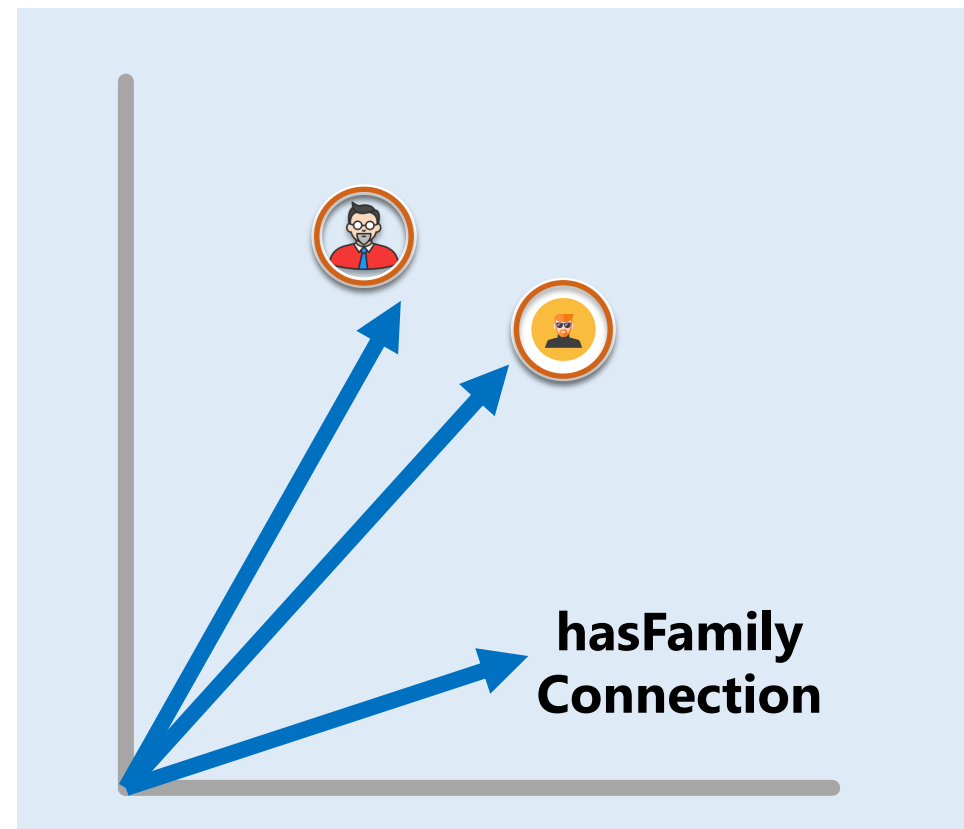
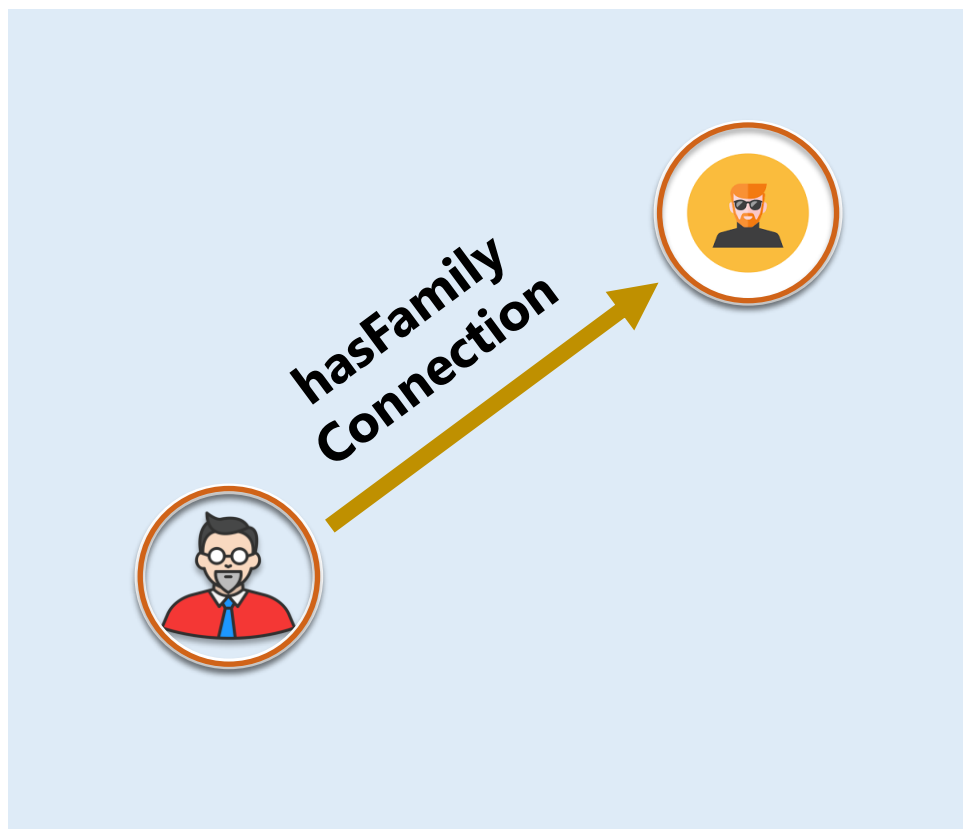


$$e: \mathbb{E} \rightarrow \mathbb{R}^2$$





$$e: (V \cup E) \rightarrow \mathbb{R}^2$$



Knowledge Graph Embeddings:

Learning **Latent** Knowledge

$$KG = (D, K; g)$$

$$K = e$$

Knowledge Graph Embeddings:

Learning **Latent** Knowledge

$$KG = (D, K, g)$$

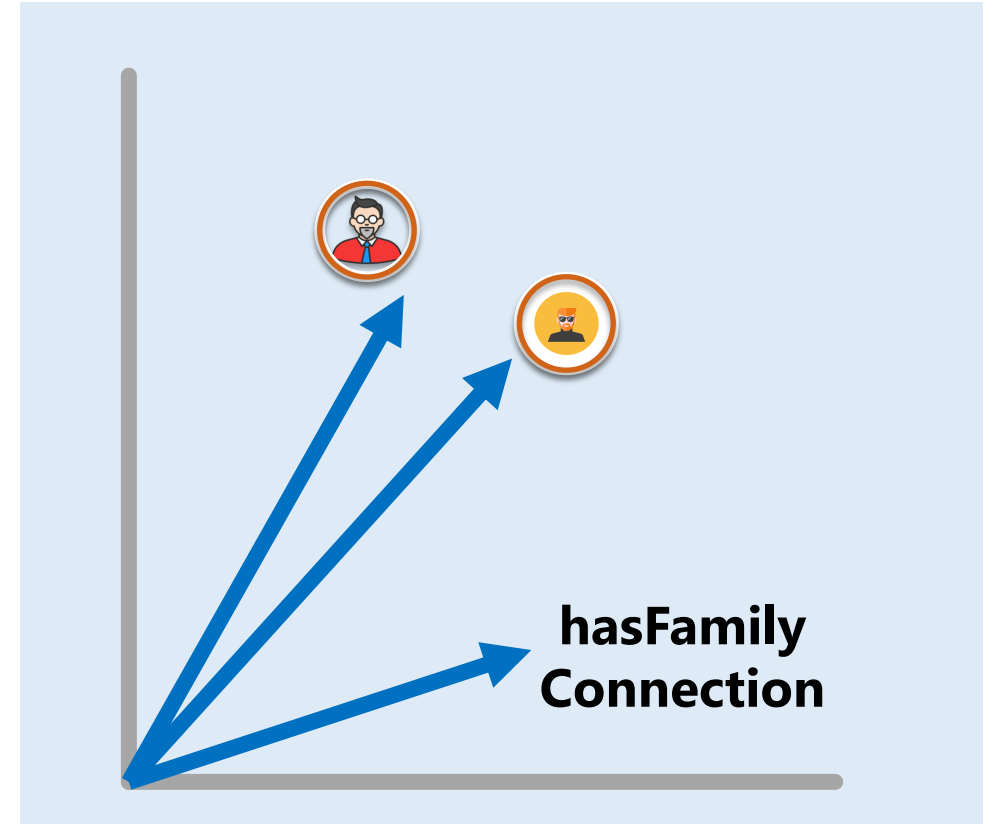
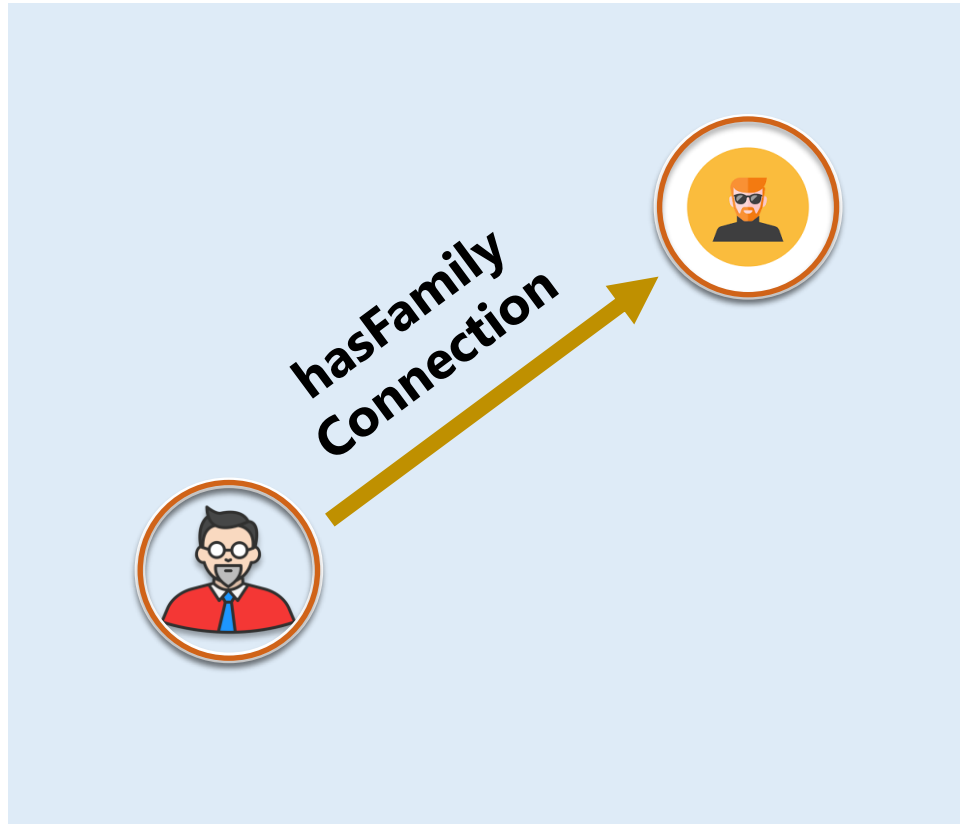
$$\mathbb{K} = \mathbb{e}$$

Knowledge Graph Embeddings:

Learning **Latent** Knowledge

$$KG = (D, K, g, l)$$

$$l: \mathbb{G} \rightarrow \mathbb{e}$$



Knowledge Graph Embeddings:

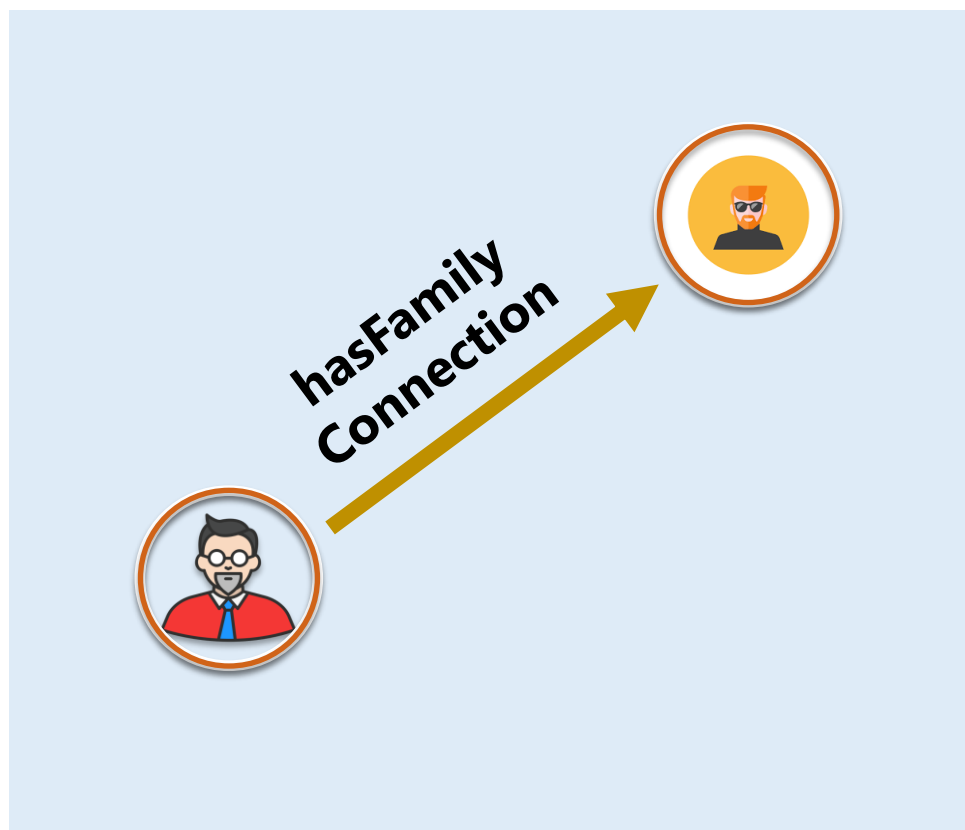
Learning **Latent** Knowledge

$$KG = (D, K; g, l)$$

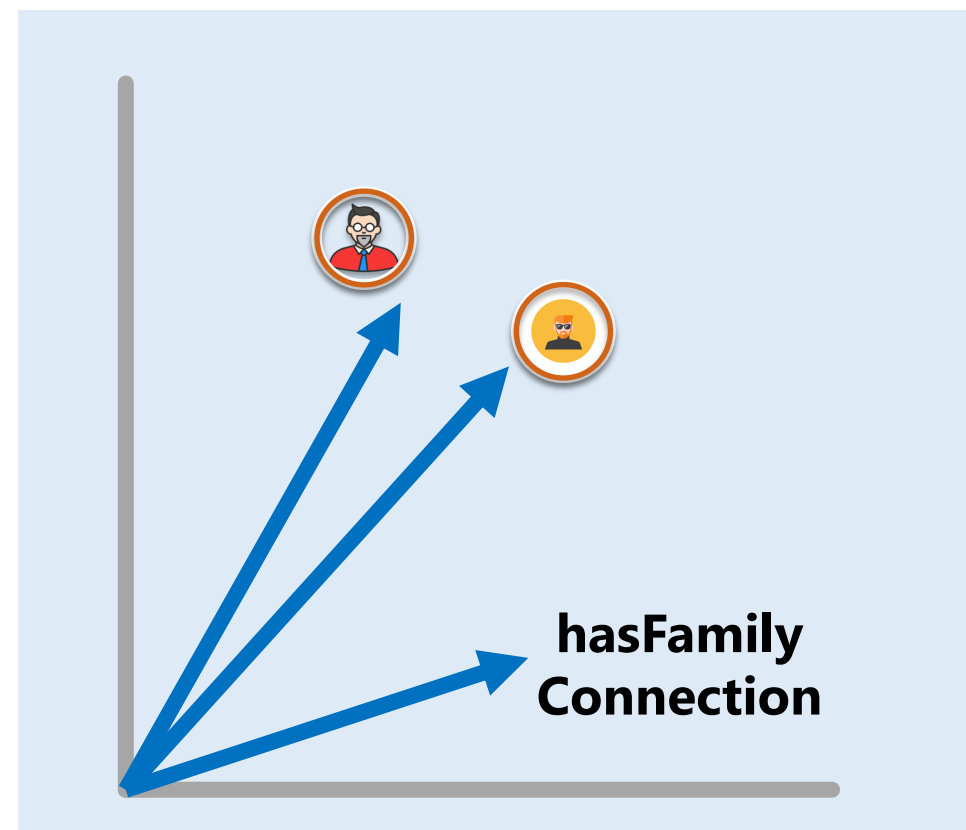
$$l: \mathbb{D} \rightarrow \mathbb{E}$$



$$l: \mathbb{D} \rightarrow \mathbb{E}$$



$\mathbb{D}$



$\mathbb{E}$

Knowledge Graph Embeddings:

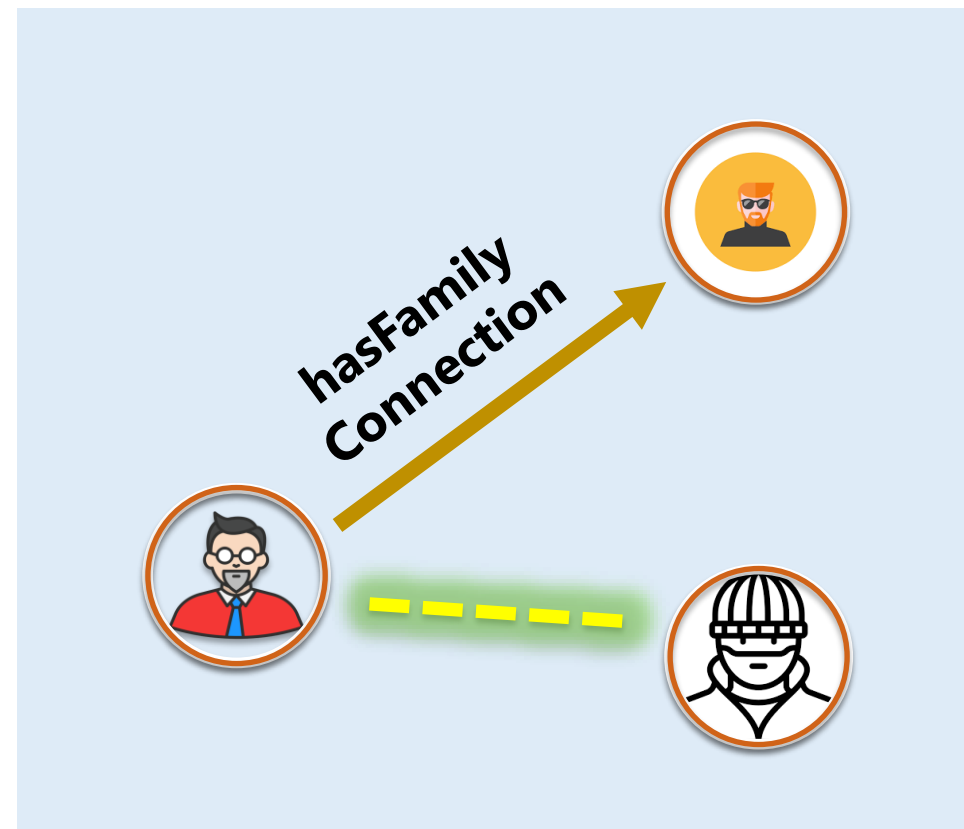
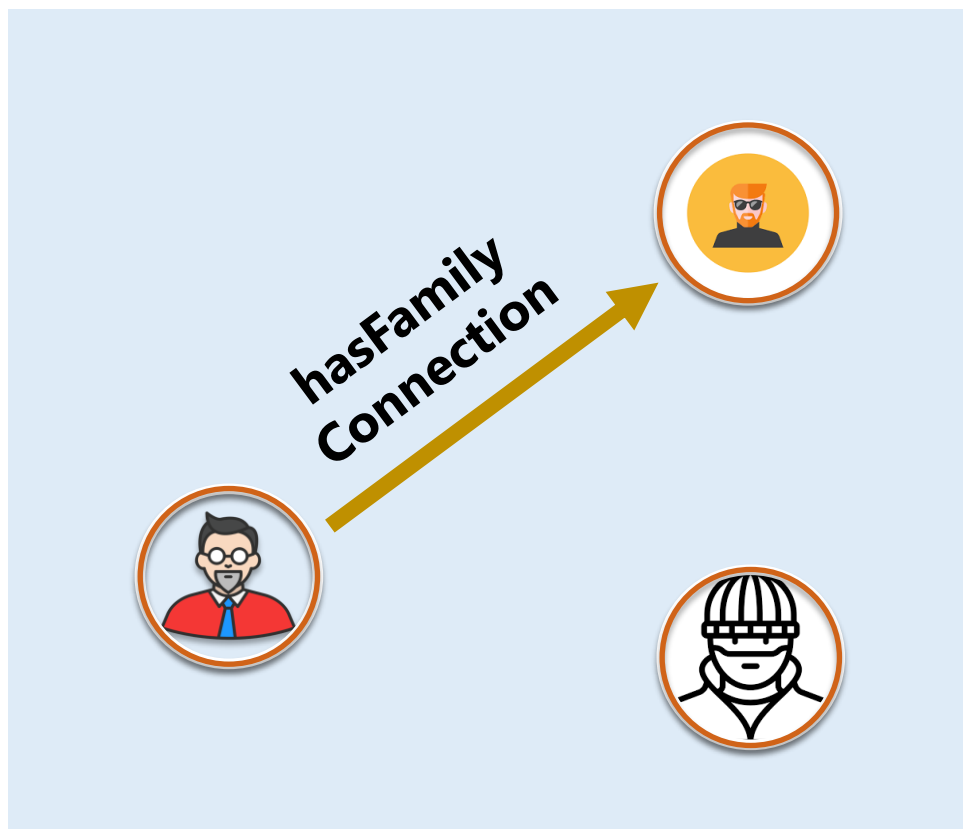
Reasoning* with **Latent** Knowledge

$$KG = (D, K; g, l, r)$$

$$r_e: \mathbb{D} \rightarrow \mathbb{D}$$

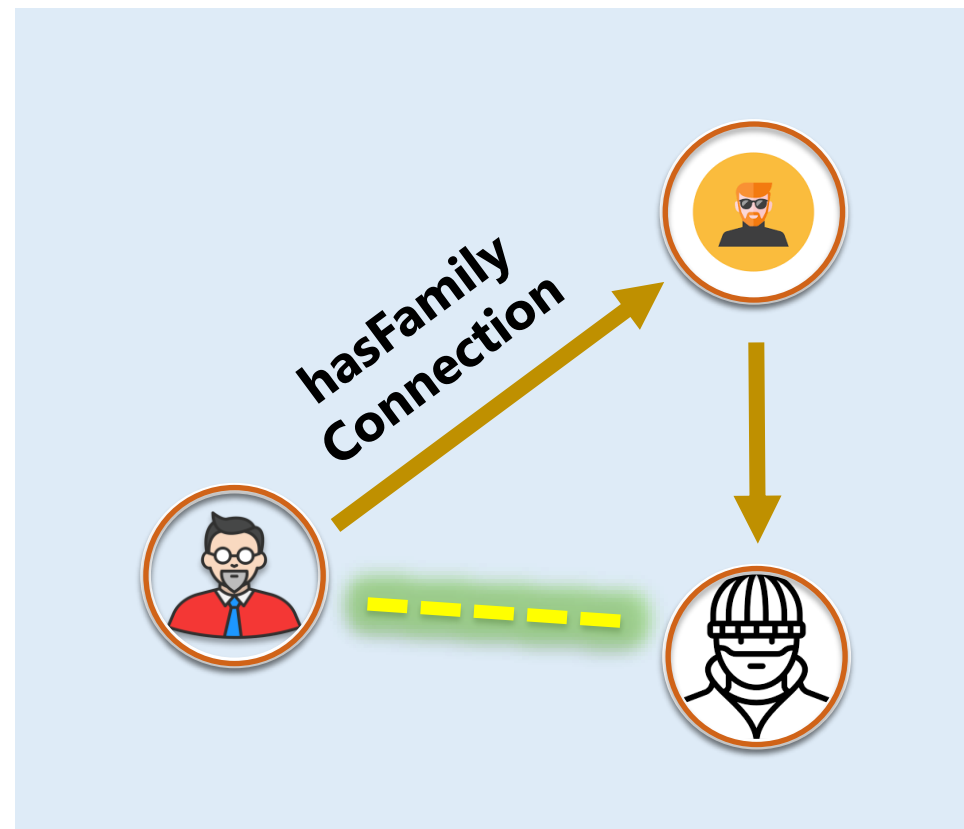
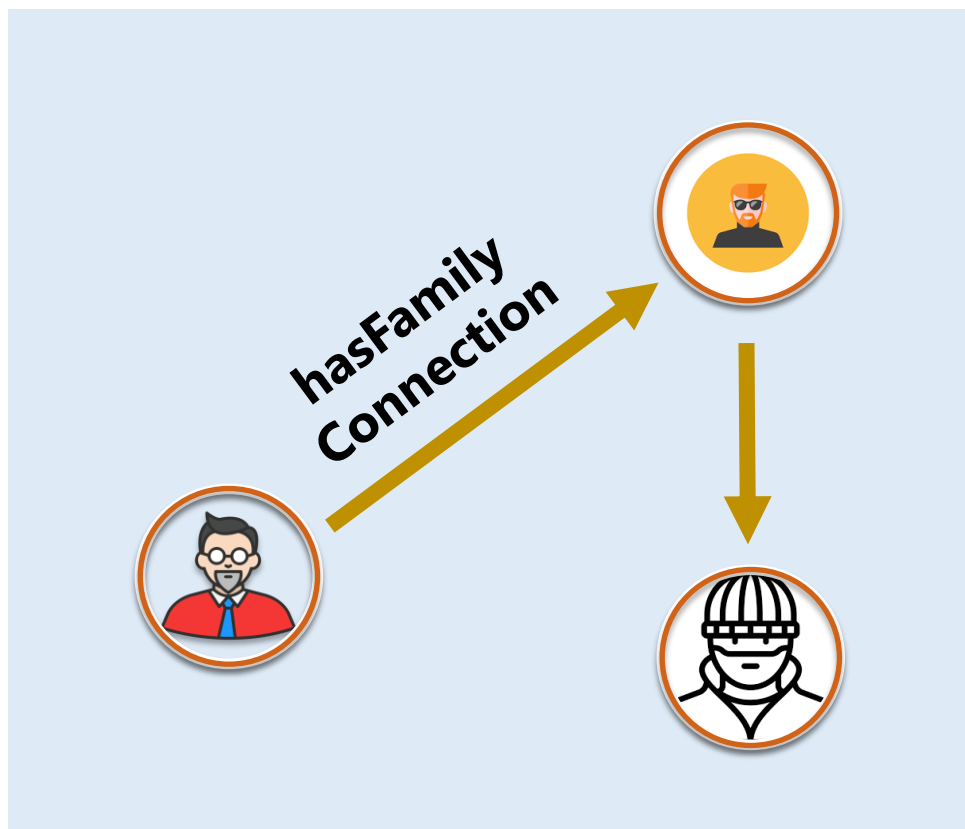


$$r_e: \mathbb{D} \rightarrow \mathbb{D}$$



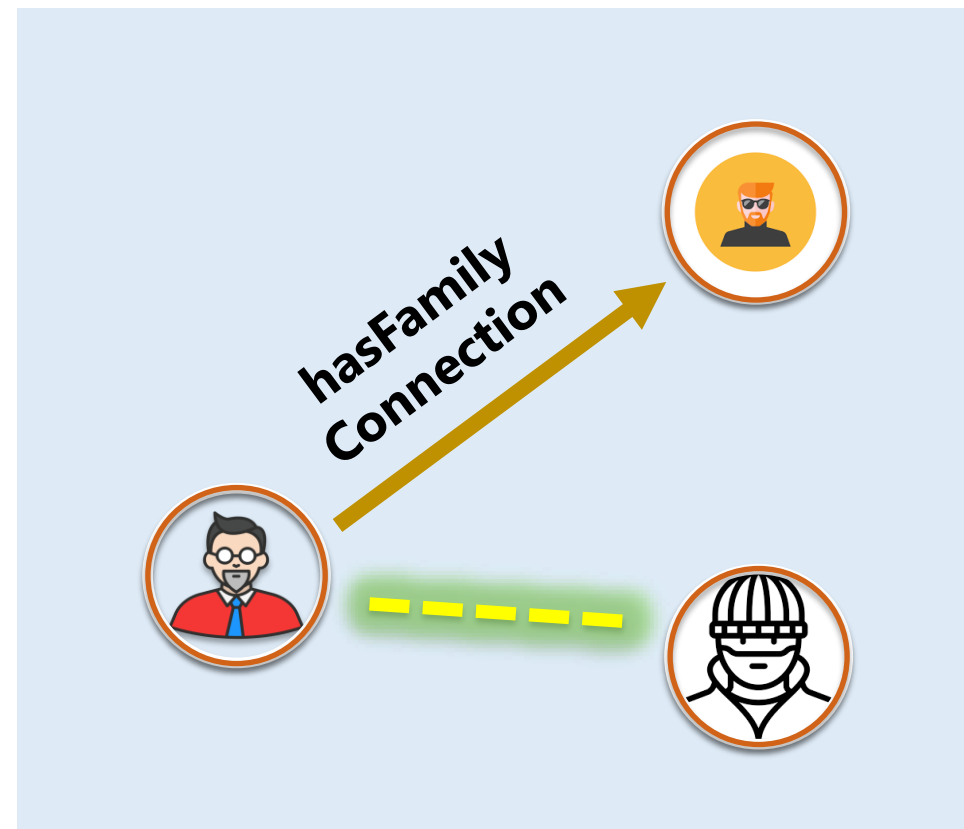
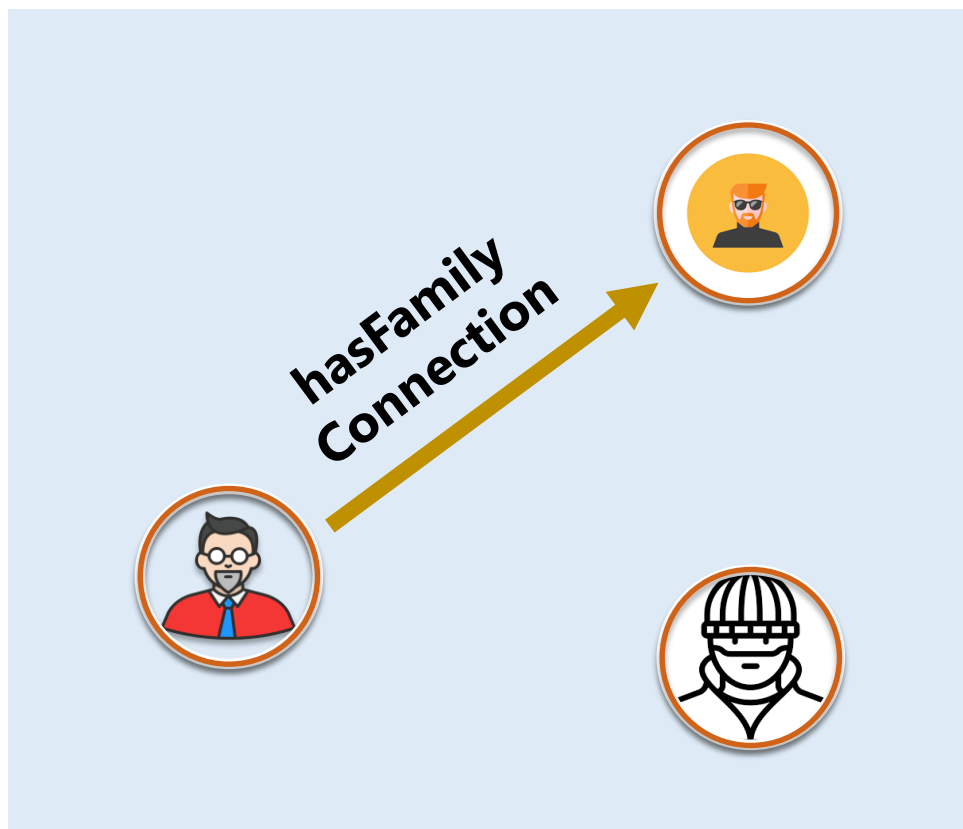


$$r_e: \mathbb{D} \rightarrow \mathbb{D}$$



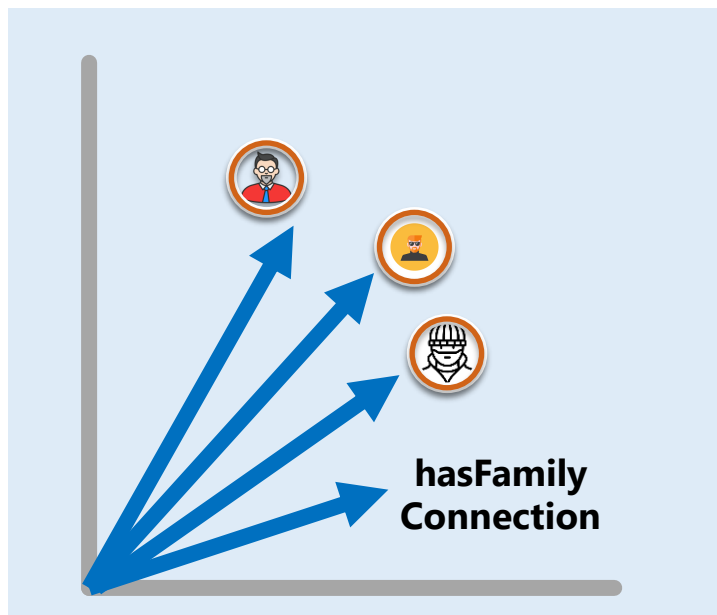


$$r_e: \mathbb{D} \rightarrow \mathbb{D}$$

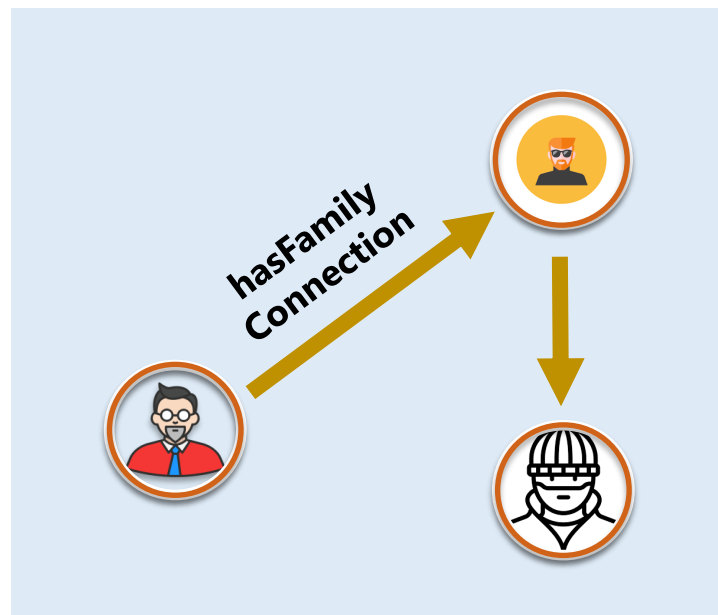




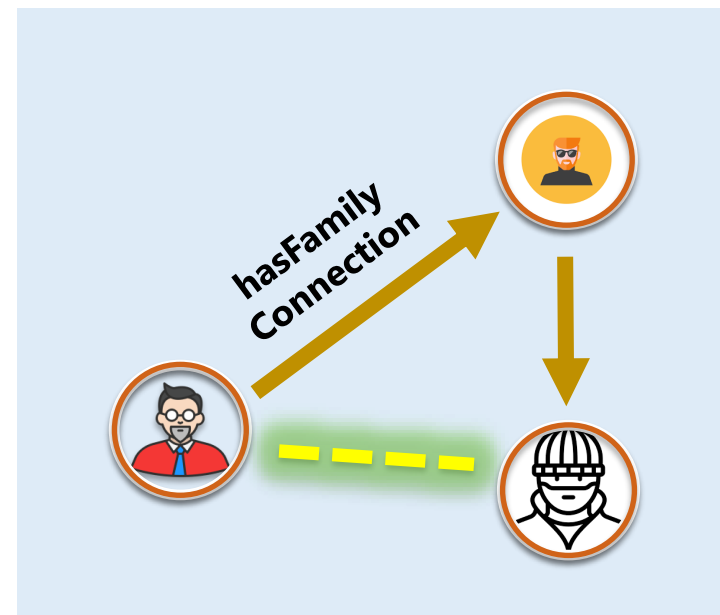
$$r_e: \mathbb{D} \rightarrow \mathbb{D}$$



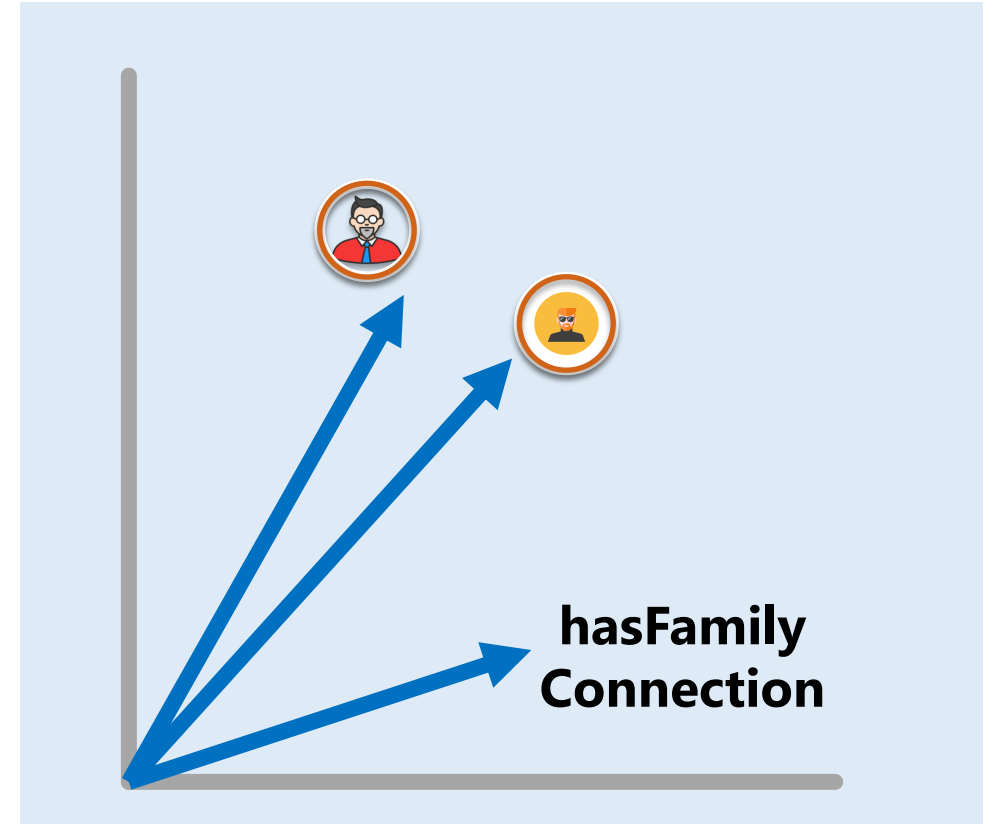
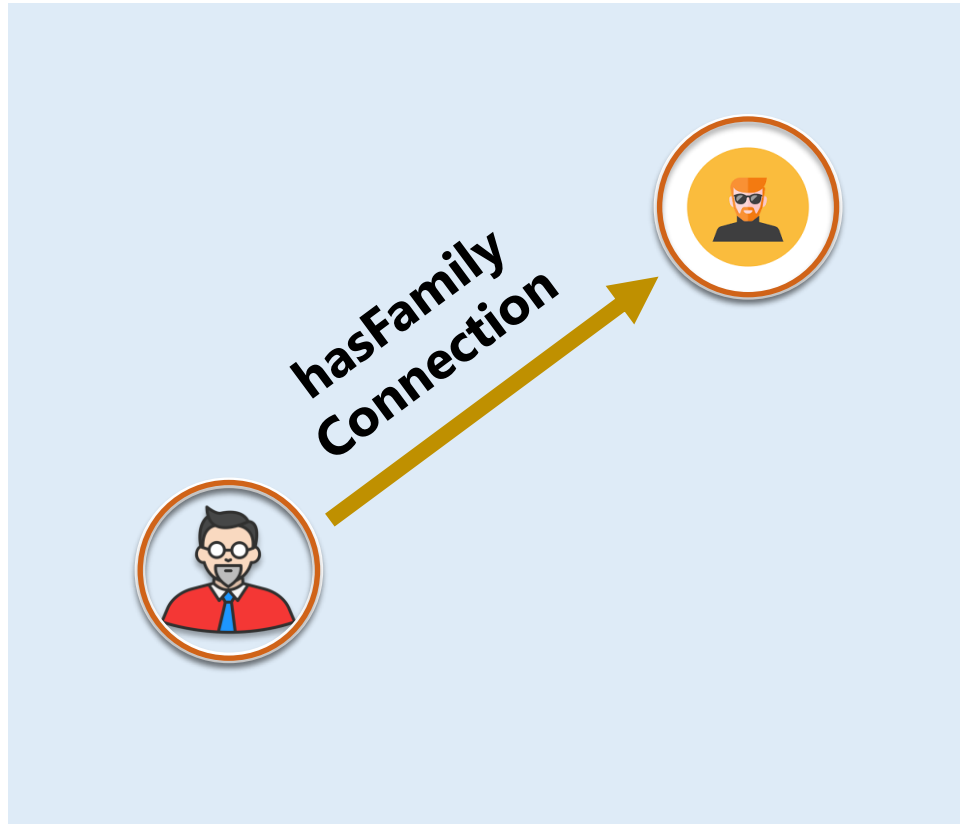
e



$\mathbb{D}$

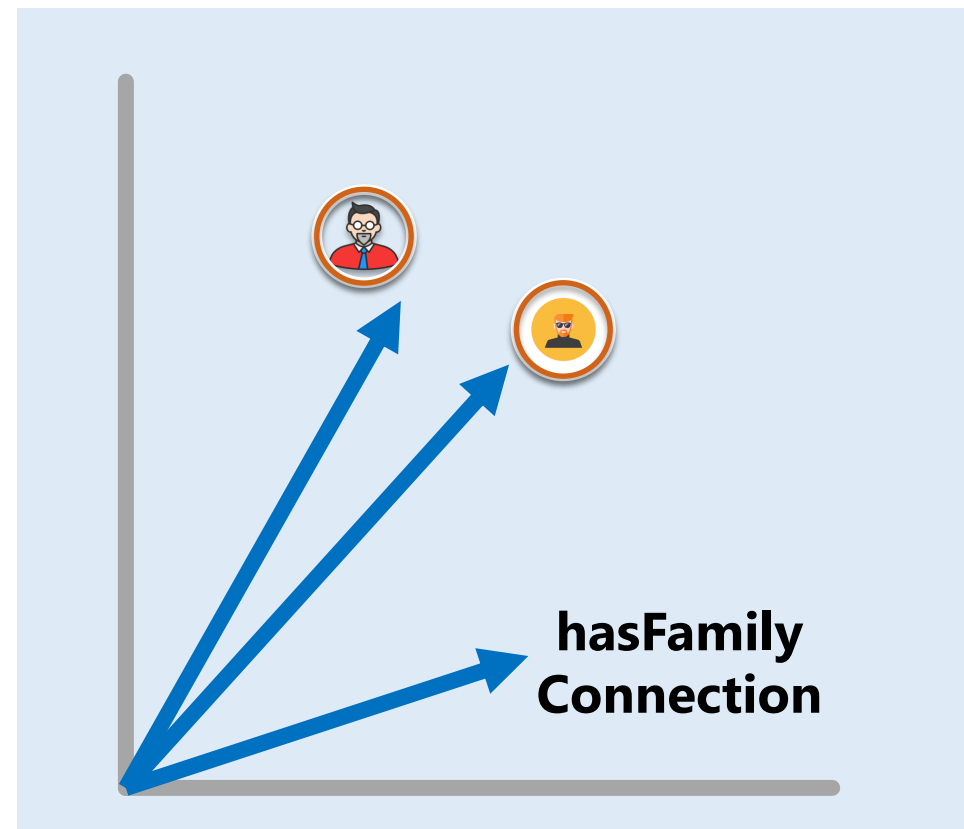
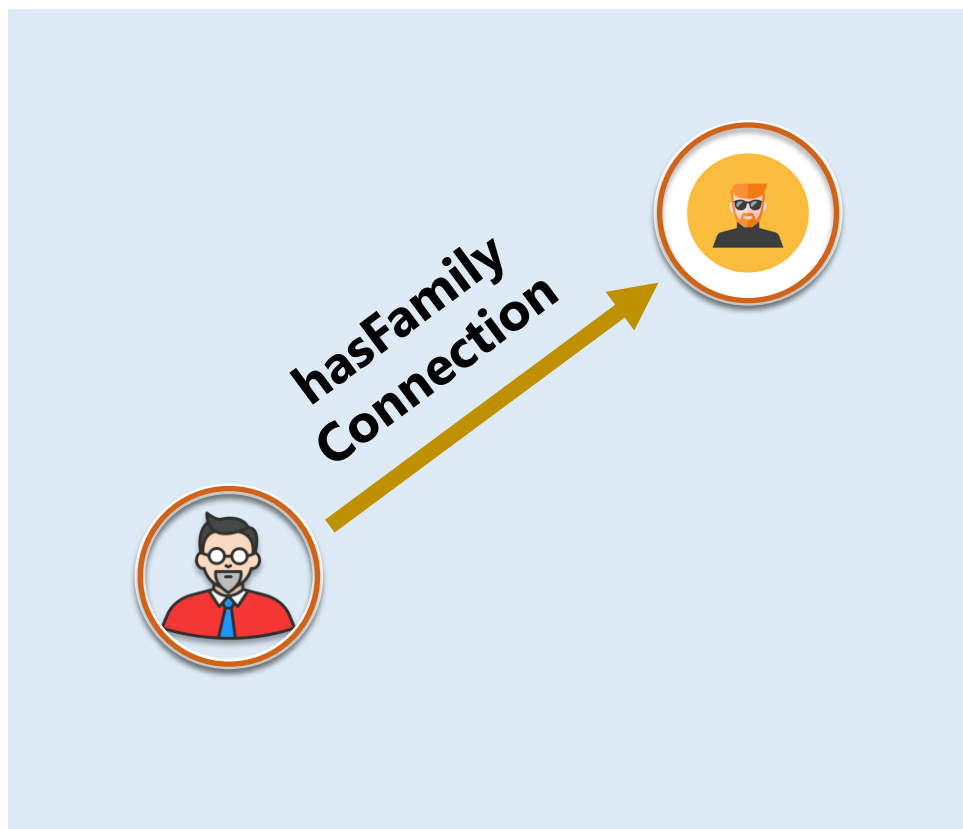


$\mathbb{D}$





$$e: (V \times E) \rightarrow \mathbb{R}^2$$



Knowledge Graph Embeddings:

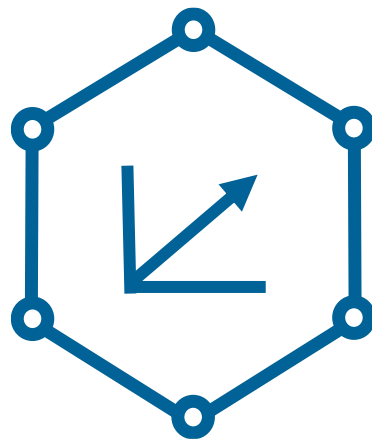
Latent Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{e}$$

$$l: \mathbb{D} \rightarrow \mathbb{e}$$

$$r_{\mathbb{e}}: \mathbb{D} \rightarrow \mathbb{D}$$



Knowledge Graph Embeddings

Emanuel Sallinger